Manan Pancholy

Scranton, PA | 570-468-2265 | U.S. Citizen | manan.pancholy@gmail.com | LinkedIn | Portfolio

Education

Brown University May 2027
ScB. Artificial Intelligence in Health Informatics (Independent Concentration) Providence, RI

- GPA: 3.96 / 4.00
- Coursework: Self-Supervised Learning (Graduate), Bioinformatics (Graduate), Advanced Deep Learning (Graduate), Machine Learning, Physiology, Biochemistry, Organic Chemistry I & II, Data Structures and Algorithms, Statistical Inference I, Linear Algebra

Professional Experience

Harvard Medical School Mar. 2026 – Present
Medical AI Researcher; PI: Pranav Rajpurkar, PhD Boston, MA

- Conceived and led AI-agent simulation of chest-pain decision-rule use in emergency department (**1st author**, under review)
- **Evaluating frontier AI models** on chest CT interpretation across **15+ countries** on 5 continents (MAIDA; program info linked here)
- Admitted through Susanne E. Churchill Summer Institute in Biomedical Informatics (≈**5% admission**)

Curricular Resource Center, Brown University Aug. 2025 – Present
Independent Concentration (IC) Co-Coordinator Providence, RI

- Advise 15+ students designing their own majors and **review 50+ proposals** a year for the College Curriculum Council
- Organized information session, proposal workshops, alumni panel, and acceptance lunch with fellow concentrators

National American Association of Physicians of Indian Origin (AAPI) Jul. 2023 – May 2025
President (prev. President-Elect), Pre-Medical Students Section Hybrid

- Co-founded the section & established annual Medical Scholars Summit **(200+ attendees)** funded by **$25,000** in sponsorships
- Provided premedical students with free Kaplan, UWorld, and Blueprint MCAT resources & seminars through corporate sponsors

Research

Selected Publications & Presentations

Pancholy MS*, Le Q*, Bhandari A, et al. Cross-Ancestry Portability of a Hypertrophic Cardiomyopathy Polygenic Score in a Diverse US Cohort. *Circ Genom Precis Med*. Published online 2026 Nov 3. doi:10.1161/CIRCGEN.126.005968 (*co-first authors)

Liu Y, Maeda K, Pancholy MS. PaCX-MAE: Physiology-Augmented Chest X-Ray Masked Autoencoder. Poster presented at: 43rd International Conference on Machine Learning (ICML) 3rd Annual Workshop on Multi-modal Foundation Models and Large Language Models for Life Sciences; 2026 Jul 11; Seoul, South Korea. arXiv:2606.01537

Pancholy MS, Dudzinski DM. Ambient Listening Artificial Intelligence: Technology and Liability Implications. Lecture presented at: American College of Cardiology Medical Professional Liability Insurance Work Group Meeting; 2024 Nov 15; Virtual.

Ongoing Work

- Deep-learning system that predicts coronary stent expansion from angiography video and clinical data (sole inventor, US Patent Application 2026/0340823 A1)
- Curating multimodal data (ECG, notes, labs) from acute coronary syndrome patients for model predicting catheterization outcomes

Certifications: HSP, GCP, COI, IRB (CITI), Advanced Catalyst Workshop

Clinical Experience

Express Urgent Care - Triage Volunteer, Medical Interpreter (Spanish, Gujarati), Check-In Clerk - 100+ patients served
Observership - 100+ hours observing primary care, GI, CT surgery, cardiology (gen, int, EP), ophthalmology, radiation oncology

Projects

AI Note-Writing | *HuggingFace Transformers, WhisperX, PyTorch, Regex, NEJM CPCs* Jun. 2024 – May 2025

- Implemented pipeline using open-source transcription + LLM to transform patient encounter audio into structured clinical notes

Brown Barsite | *CSS, JavaScript, HTML, Adobe Lightroom* Jun. 2024 – Present

- Developed & maintain multi-page responsive website for Brown's South Asian fusion a cappella group (while Music Director & Bass)

Technical Skills

Languages: Python, Julia, C/C++, R, Stata, Java, SQL, HTML, CSS, JavaScript
Libraries: PyTorch (Base, Lightning, Geometric), TensorFlow, Transformers, RDKit, Scikit-learn, NumPy
Developer Tools: VS Code, GitHub, Slurm (HPC clusters), WandB, CUDA (DDP), Docker
Interests: Voice impressions (Simpsons, iPhone), beatboxing, cooking, hiking, menswear, Studio Ghibli, *Tuesdays with Morrie*